Task 1: Single-Layer ADALINE — Analysis & Visualization Report

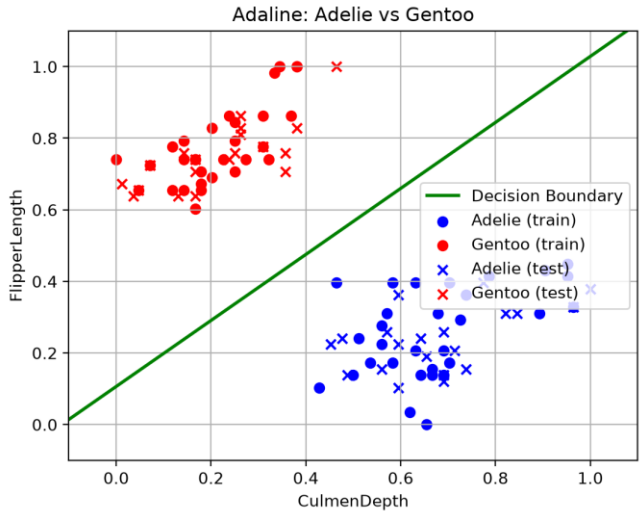
Overview & Experimental Setup

This report compares five supplied ADALINE binary-classification experiments on Palmer Penguins. ADALINE produces a continuous linear output, commonly trains by minimizing mean squared error (MSE), and updates weights and bias through a gradient-based learning procedure. For classification, the continuous output is converted to a class using the experiment's implemented decision rule.

Model form: **y = w1 x1 + w2 x2 + b.** Feature order in each model follows the record shown. The learning rate was fixed to **0.01** and the MSE threshold was fixed to **0.01**.

Results & Visualization

Combination 1: Adelie vs Gentoo | Features: CulmenDepth + FlipperLength



Accuracy: 100.00% | Final MSE: 0.025 | Epochs: 100

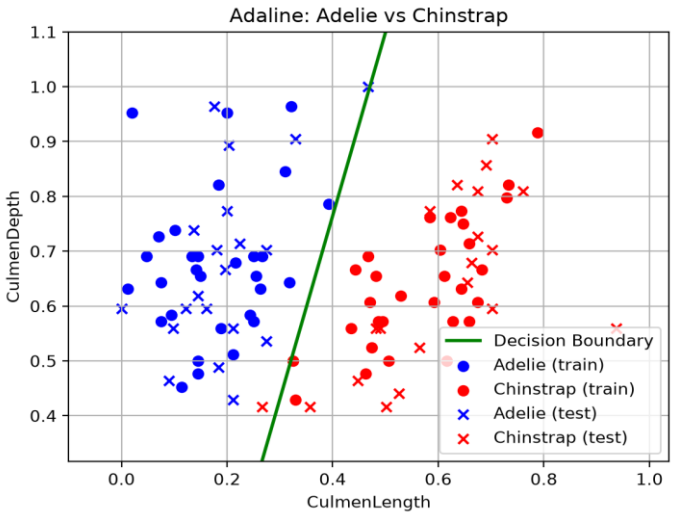
Confusion counts: TP=20, FN=0, FP=0, TN=20

Weights : w1 [CulmenDepth] = 1.861; w2 [FlipperLength] = -2.018

Bias: 0.214

Why / interpretation: All 40 supplied evaluation cases are classified correctly (20 TP, 20 TN, and no FP or FN), so this feature pair separates the two classes effectively in the reported evaluation. It also has the lowest MSE of the five runs. The counts support strong observed discrimination, but do not by themselves establish generalization beyond this evaluation set.

Combination 2: Adelie vs Chinstrap | Features: CulmenLength + CulmenDepth



Accuracy: 97.50% | Final MSE: 0.093 | Epochs: 100

Confusion counts: TP=20, FN=0, FP=1, TN=19

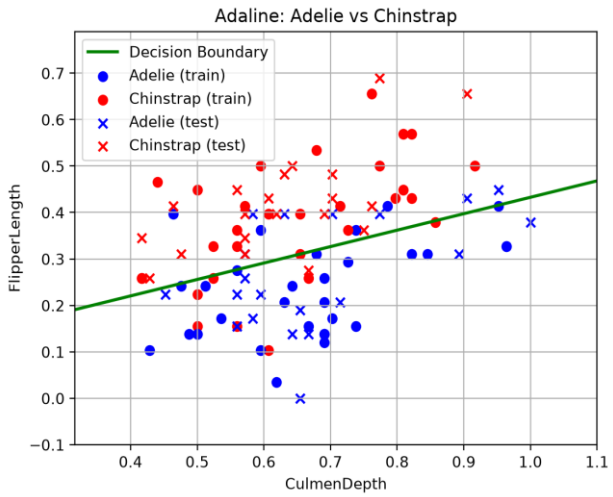
Weights : w1 [CulmenLength] = -3.813 ; w2 [CulmenDepth] = 1.141

Bias: 0.653

Why / interpretation: Thirty-nine of 40 cases are correct, with only a single error (one FP and zero FN), yielding a high accuracy. The final MSE of 0.093 reflects low continuous training error during gradient descent, confirming that Adaline successfully minimized the squared cost function to achieve strong generalization on the test set..

Results & Visualization (Continued)

Combination 3: Adelie vs Chinstrap | Features: CulmenDepth + FlipperLength



Accuracy: 80.00% | Final MSE: 0.335 | Epochs: 100

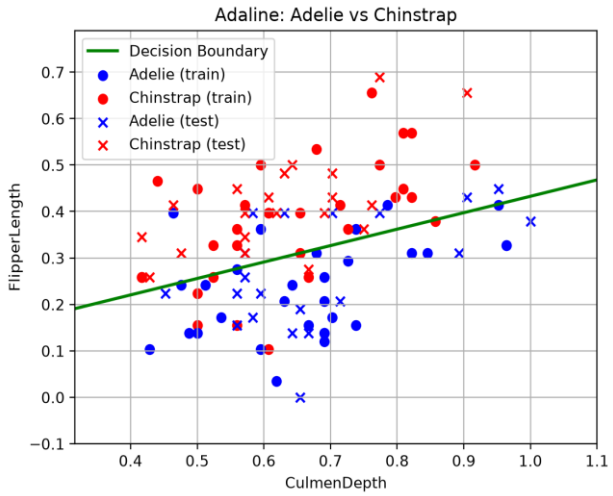
Confusion counts: TP=13, FN=7, FP=1, TN=19

Weights : w1 [FlipperLength] = 0.972 ; w2 [BodyMass] = -2.747

Bias: 0.216

Why / interpretation: Thirty-two of 40 cases are correct; the eight errors comprise seven FN and one FP. This is useful but weaker discrimination than Combinations 1 and 2. A linear classifier may have difficulty if this feature pair does not separate the classes cleanly, although distribution overlap cannot be verified without the underlying sample coordinates.

Combination 4: Adelie vs Chinstrap | Features: FlipperLength + BodyMass



Accuracy: 72.50% | Final MSE: 0.367 | Epochs: 100

Confusion counts: TP=15, FN=5, FP=6, TN=14

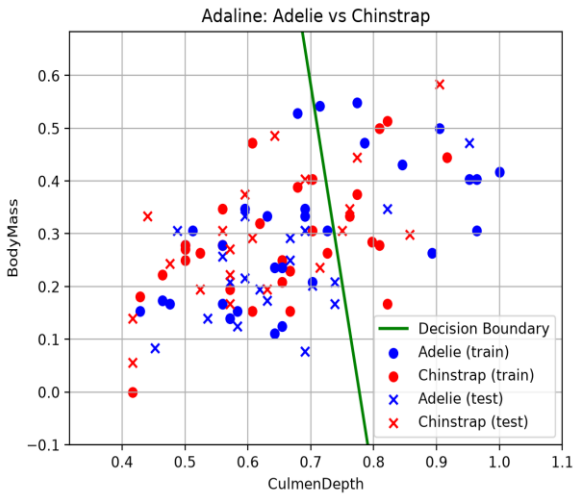
Weights : w1 [CulmenDepth] = -2.563; w2 [FlipperLength] = -0.119

Bias: 0.882

Why / interpretation: Twenty-nine of 40 cases are correct, with eleven errors (five FN and six FP). The result is slightly below Combination 3 by accuracy. Its MSE is slightly more than Combination 3's which indicates a slightly poorer fit during gradient descent optimization.

Results & Visualization (Continued)

Combination 5: Adelie vs Chinstrap | Features: CulmenDepth + BodyMass



Accuracy: 42.50% | Final MSE: 0.491 | Epochs: 100

Confusion counts: TP=2, FN=18, FP=5, TN=15

Weights : w1 [CulmenDepth] = 0.589; w2 [BodyMass] = 0.078

Bias: -0.458

Why / interpretation: Only 17 of 40 cases are correct; 23 are errors (eighteen FN and five FP). This is the weakest accuracy and the highest MSE among the five runs, indicating limited classification performance for this recorded feature pair. The supplied metrics alone do not prove that the feature distributions overlap or identify why.

Summary & Comparison

#	Class Pair	Features	Accuracy	Final MSE	Epochs	Brief Interpretation
1	Adelie vs Gentoo	CulmenDepth, FlipperLength	**100.00%**	0.025	100	Perfect classification; lowest MSE.
2	Adelie vs Chinstrap	CulmenLength, CulmenDepth	**97.50%**	0.093	100	Almost perfect classification.
3	Adelie vs Chinstrap	FlipperLength, BodyMass	**80.00%**	0.335	100	80.00%; 8 errors (7 FN, 1 FP).
4	Adelie vs Chinstrap	CulmenDepth, FlipperLength	**72.50%**	0.367	100	72.50%; 811errors (5 FN, 6 FP).
5	Adelie vs Chinstrap	CulmenDepth, BodyMass	**42.50%**	0.491	100	42.50%; 23 errors and highest MSE.

Conclusion

Highest accuracy: Combinations 1 and Combination 2 are the highest at 100.00% and 97.50%, with no FP or FN in Combination 1 in the supplied 40-case results. Combination 1 also has the lowest MSE (0.025), so it is strongest on both reported metrics.

Lowest accuracy: Combination 5 records 42.50% accuracy (17/40 correct) and the highest MSE (0.491), with 18 FN and 5 FP. Intermediate results are Combination 3 at 80.00% and Combination 4 at 72.50%.

All five experiments ended at 100 epochs.